

ADVANCED PROGRAMMING SPRING 26

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ASSIGNMENT 1

Q. Write a C program to analyze time complexities in constant time, linear time and quadratic time. Increase the size of input to check the time consumed



```
#include <stdio.h>
```

```
#include <time.h>
```

```
void constantTime(int arr[]) {
```

```
    int x = arr[0];    // single operation (O(1))
```

```
}
```

```
void linearTime(int arr[], int n) {
```

```
    int sum = 0;
```

```
    for(int i = 0; i < n; i++) {    // runs n times (O(n))
```

```
        sum += arr[i];
```

```
    }
```

```
}
```

```
void quadraticTime(int arr[], int n) {
```

```
    int count = 0;
```

```
    for(int i = 0; i < n; i++) {    // nested loops (O(n^2))
```

```
        for(int j = 0; j < n; j++) {
```

```
            count++;
```

```
        }
```

```
}  
}
```

```
int main() {  
    int sizes[] = {100, 500, 1000, 2000};  
    int totalTests = 4;  
  
    printf("Input Size | O(1) Time | O(n) Time | O(n^2) Time\n");  
    printf("-----\n");  
  
    for(int s = 0; s < totalTests; s++) {  
        int n = sizes[s];  
        int arr[n];  
  
        for(int i = 0; i < n; i++) {  
            arr[i] = i;  
        }  
  
        clock_t start, end;  
  
        start = clock();  
        constantTime(arr);  
        end = clock();  
        double t1 = (double)(end - start) / CLOCKS_PER_SEC;  
  
        start = clock();  
        linearTime(arr, n);  
        end = clock();
```

```

double t2 = (double)(end - start) / CLOCKS_PER_SEC;

start = clock();
quadraticTime(arr, n);
end = clock();
double t3 = (double)(end - start) / CLOCKS_PER_SEC;

printf("%10d | %0.6f | %0.6f | %0.6f\n", n, t1, t2, t3);
}

return 0;
}

```

Output:

Input Size | O(1) Time | O(n) Time | O(n²) Time

```

100 | 0.000001 | 0.000000 | 0.000006
500 | 0.000000 | 0.000001 | 0.000147
1000 | 0.000000 | 0.000003 | 0.000463
2000 | 0.000000 | 0.000004 | 0.001610

```